Comptia N+ Assignment

[N+ Network Fundamental and building networks]

Introduction to IP Addressing

IPv4 and IPv6. Differences, structure, and importance. Subnetting basics: CIDR, subnet masks. Assign static IPs to devices and verify connectivity.

Task: Solve subnetting exercises and calculate valid subnets.

❖ **1. List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.**

➢ **Aim**

- To understand IP addressing and identify the key differences between IPv4 and IPv6 in terms of address structure, size, configuration, and importance.

➢ **Introduction**

- An **IP (Internet Protocol) address** is a logical address assigned to a device connected to a network. It helps identify the device and enables data packets to travel from a source to a destination across networks.
- There are two main versions of IP addressing: **IPv4 (Internet Protocol version 4)** and **IPv6 (Internet Protocol version 6)**. IPv4 is the older and widely used version, while IPv6 was developed to provide a much larger address space and improve the scalability of Internet communication.

➢ Key Differences Between IPv4 and IPv6

IPv4	IPv6
1. Address Size: IPv4 uses a 32-bit address.	**1. Address Size:** IPv6 uses a 128-bit address.
2. Address Space: It provides approximately 4.3 billion possible addresses.	**2. Address Space:** It provides an extremely large number of addresses, approximately (3.4 \times 10^{38}).
3. Address Format: It is written in four decimal numbers separated by dots. Example: 192.168.1.10	**3. Address Format:** It is written in hexadecimal groups separated by colons. Example: 2001:db8::10
4. Configuration: Addresses can be assigned manually or automatically using DHCP.	**4. Configuration:** Addresses can be assigned manually, through DHCPv6, or automatically using SLAAC.
5. Header Structure: IPv4 has a variable-length header of 20–60 bytes.	**5. Header Structure:** IPv6 has a fixed 40-byte base header.
6. Broadcast: IPv4 supports broadcast communication.	**6. Broadcast:** IPv6 does not use broadcast; it uses multicast and anycast for relevant communication.
7. NAT Usage: NAT is commonly used to allow multiple devices to share a public IPv4 address.	**7. NAT Usage:** IPv6 provides a very large address space, reducing the need for address-sharing NAT, although translation can still be used.

8. Security: IPsec support is available for IPv4, but its use depends on the implementation and configuration.	**8. Security:** IPv6 includes standardized support for IPsec, but encryption is not automatically enabled for all traffic.

- **Importance of IPv4 and IPv6**
- Both IPv4 and IPv6 are important for communication between devices over networks.
  - **IPv4** remains widely used and supports many existing networks and devices.
  - **IPv6** provides a much larger address space to support the growing number of Internet-connected devices.
  - Both protocols help route data packets from the source device to the destination.
  - Understanding both versions is important for network configuration, troubleshooting, and subnetting.

❖ **2. Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.**

➢ **Aim**

- To assign unique static IPv4 addresses to four home Wi-Fi devices using the 192.168.10.0/24 network and identify the subnet mask and default gateway for each device.

➢ **Introduction**

➢ An **IP address** is a logical address used to identify a device on a network. A static IP address is manually configured and remains the same unless it is changed by the user or network administrator.

➢ In a home network, devices such as laptops, mobile phones, smart TVs, and printers require unique IP addresses to communicate with each other and access the Internet.

➢ The given network is 192.168.10.0/24. The /24 prefix means that the first 24 bits represent the network portion of the address.

- **Network Address:** 192.168.10.0
- **Subnet Mask:** 255.255.255.0
- **Usable Host Range:** 192.168.10.1 to 192.168.10.254
- **Broadcast Address:** 192.168.10.255
- **Default Gateway:** 192.168.10.1 (assumed home router address)

➢ **Static IP Address Assignment Table**

➢ The following table assigns a unique static IPv4 address to each device.

Device	IP Address	Subnet Mask	Default Gateway
Laptop	192.168.10.10	255.255.255.0	192.168.10.1
Phone	192.168.10.11	255.255.255.0	192.168.10.1
Smart TV	192.168.10.12	255.255.255.0	192.168.10.1
Printer	192.168.10.13	255.255.255.0	192.168.10.1

- **Explanation**
- **1. Laptop — 192.168.10.10**
  - The laptop is assigned the IP address 192.168.10.10. It can communicate with other devices on the same network and access the Internet through the default gateway.
- **2. Phone — 192.168.10.11**
  - The phone is assigned the IP address 192.168.10.11. This address is different from the laptop's address, ensuring that both devices can be identified separately.
- **3. Smart TV — 192.168.10.12**
  - The smart TV is assigned the IP address 192.168.10.12. It can communicate with the home router to access online streaming services.
- **4. Printer — 192.168.10.13**
  - The printer is assigned the IP address 192.168.10.13. Other devices on the same network can use this address to communicate with the printer.

- **Subnet Mask**
  - All four devices use the subnet mask 255.255.255.0, which corresponds to /24. It indicates that the devices belong to the same IP network.

- **Default Gateway**
  - The default gateway is 192.168.10.1. It represents the home router, which forwards traffic from the local network to other networks, such as the Internet.

- **Important Configuration Note**
  - These are example static IP assignments. Before configuring the devices, verify that the router uses 192.168.10.1 as its address and ensure that the selected IP addresses are not already assigned to other devices or included in a conflicting DHCP allocation.

- **Result**
  - Four devices—laptop, phone, smart TV, and printer—were assigned unique static IPv4 addresses within the 192.168.10.0/24 network. Each device uses the same subnet mask and default gateway.

❖ **3. Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.**

➢ **Aim**

- To calculate the number of valid subnets and usable hosts per subnet for the 192.168.1.0 network using a /27 CIDR prefix and identify the first three subnet ranges.

➢ **Introduction**

- Subnetting is the process of dividing a large IP network into smaller networks called subnets. It improves IP address management, reduces unnecessary broadcast traffic, and helps organize devices efficiently.
- CIDR stands for **Classless Inter-Domain Routing**. A /27 prefix means that 27 bits are used for the network portion of the IPv4 address, while the remaining bits are used for host addresses.

➢ **Given Information**

- **Network:** 192.168.1.0
- **CIDR Prefix:** /27
- **IPv4 Address Size:** 32 bits
- **Default Class C Prefix:** /24

➢ **Calculation of Subnets and Hosts**

➢ **A. Number of Host Bits**

➢ The number of host bits is calculated as:

- Host bits = 32 − Prefix length
- Host bits = 32 − 27 = **5 bits**

- **B. Number of Subnets**
- Starting with a /24 network and borrowing 3 bits gives a /27 prefix.
- Borrowed bits = 27 − 24 = 3 bits
- Number of subnets = (2^3 = 8)
- Therefore, **8 subnets** are available when dividing the original /24 network into /27 subnets.

- **C. Number of Hosts per Subnet**
- Number of addresses per subnet = (2^5 = 32)
- Usable hosts per subnet = (2^5 - 2 = 30)
- Two addresses are reserved in each subnet: one for the network address and one for the broadcast address.
- Therefore, **each subnet has 30 usable host IP addresses**.

- **D. Subnet Mask**
- The /27 CIDR prefix corresponds to the subnet mask:
- **255.255.255.224**

- **E. Block Size**
- Block size = 256 − 224 = **32**
- Therefore, the subnet network addresses increase in steps of 32 in the last octet.

- **First Three Valid Subnet Ranges**

Subnet	Network Address	First Usable Host	Last Usable Host	Broadcast Address
1	192.168.1.1	192.168.1.1	192.168.1.30	192.168.1.31
2	192.168.1.32	192.168.1.33	192.168.1.62	192.168.1.63
3	192.168.1.64	192.168.1.65	192.168.1.94	192.168.1.95

- **Address Ranges Explained**
- **Subnet 1: 192.168.1.0/27**
  - Network address: 192.168.1.0
  - Usable host range: 192.168.1.1 to 192.168.1.30
  - Broadcast address: 192.168.1.31

- **Subnet 2: 192.168.1.32/27**
  - Network address: 192.168.1.32
  - Usable host range: 192.168.1.33 to 192.168.1.62
  - Broadcast address: 192.168.1.63

- **Subnet 3: 192.168.1.64/27**
  - Network address: 192.168.1.64
  - Usable host range: 192.168.1.65 to 192.168.1.94
  - Broadcast address: 192.168.1.95

- **Final Answer**
  - **Total number of /27 subnets:** 8

- **Total IP addresses per subnet:** 32
- **Usable hosts per subnet:** 30
- **CIDR prefix:** /27
- **Subnet mask:** 255.255.255.224
- **Block size:** 32

➢ The first three subnet ranges are 192.168.1.0–192.168.1.31, 192.168.1.32–192.168.1.63, and 192.168.1.64–192.168.1.95. The usable host ranges are 192.168.1.1–192.168.1.30, 192.168.1.33–192.168.1.62, and 192.168.1.65–192.168.1.94, respectively.

❖ **4. You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet. Convert the subnet mask to CIDR notation and use bitwise AND for network address calculation.**

➢ **Aim**

- To calculate the network address, broadcast address, and valid host address range for a given IPv4 address using a subnet mask and the bitwise AND operation.

➢ **Given Information**

- **IP Address:** 10.0.5.17
- **Subnet Mask:** 255.255.255.0

➢ **Conversion of Subnet Mask to CIDR Notation**

- The subnet mask is:
- 255.255.255.0
- In binary, it is represented as:
- 11111111.11111111.11111111.00000000
- The subnet mask contains 24 consecutive 1 bits and 8 0 bits.
- Therefore, the CIDR notation is **/24**.
- The network is written as:
- 10.0.5.17/24

➢ **Calculation of Network Address Using Bitwise AND**

➢ The network address is calculated by performing a bitwise AND operation between the IP address and the subnet mask.

- The AND operation follows these rules:

Input A	Input B	Output
0	0	0
0	1	0
1	0	0
1	1	1

- **Step 1: Convert the IP Address to Binary**
  - IP Address: 10.0.5.17
  - Binary representation:
  - 00001010.00000000.00000101.00010001

- **Step 2: Convert the Subnet Mask to Binary**
  - Subnet Mask: 255.255.255.0
  - Binary representation:
  - 11111111.11111111.11111111.00000000

- **Step 3: Perform the Bitwise AND Operation**
- IP Address: 00001010.00000000.00000101.00010001
- Subnet Mask: 11111111.11111111.11111111.00000000
- ---
- Network: 00001010.00000000.00000101.00000000
  - Converting the result back to decimal gives:
  - **Network Address: 10.0.5.0**

- **Calculation of Broadcast Address**
  - The broadcast address is obtained by setting all host bits to 1.
  - For a /24 subnet, the last 8 bits are the host portion.
  - Network Address: 10.0.5.0
  - Broadcast Address: 10.0.5.255
  - Therefore, **Broadcast Address = 10.0.5.255**.

- **Range of Valid Host Addresses**
- The network address and broadcast address cannot normally be assigned to individual hosts in this subnet.
  - **First Valid Host:** 10.0.5.1
  - **Last Valid Host:** 10.0.5.254
  - **Total IP Addresses:** (2^8 = 256)
  - **Usable Host Addresses:** (2^8 - 2 = 254)
- Therefore, the valid host range is:
  - **10.0.5.1 to 10.0.5.254**

- **Final Answer**

Particular	**Answer**
Given IP Address	10.0.5.17
Subnet Mask	255.255.255.0
CIDR Notation	/24

Network Address	10.0.5.0
Broadcast Address	10.5.5.255
First Valid Host	10.0.5.1
Last Valid Host	10.0.5.254
Total IP Addresses	256
Usable Host Addresses	254

❖ **5. Use an online subnet calculator tool (such as https://www.subnet-calculator.com/) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.**

➢ **Aim**

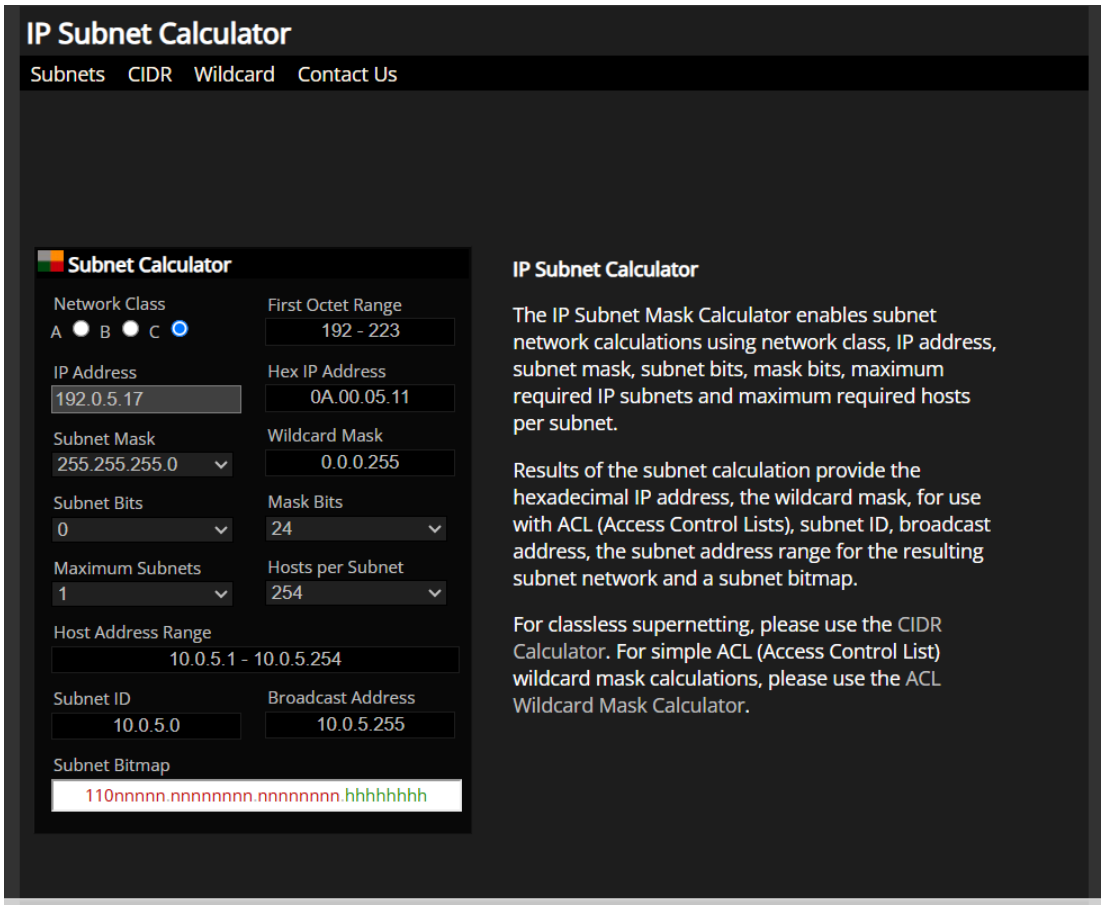
- To verify the manually calculated network address, broadcast address, and valid host address range using an online subnet calculator.

➢ **Introduction**

- An online subnet calculator is a networking tool used to calculate network details from an IP address and subnet mask. It helps verify the network address, broadcast address, CIDR notation, and valid host range obtained through manual calculations.

➢ **Procedure**

1. Open an online subnet calculator: https://www.subnet-calculator.com/
2. Enter the IP address: **10.0.5.17**
3. Enter the subnet mask: **255.255.255.0**
4. Click the Calculate button if required.
5. Check the network address, broadcast address, CIDR notation, and valid host range displayed by the tool.
6. Take a screenshot of the calculator showing the entered values and calculated results, and attach it below.



➢ Verification Results

Parameter	Manual Calculation	Calculator Output
IP Address	10.0.5.17	10.0.5.17
Subnet Mask	255.255.255.0	255.255.255.0

CIDR Notation	/24	/24
Network Address	10.0.5.0	10.0.5.0
Broadcast Address	10.0.5.255	10.0.5.255
First Valid Host	10.0.5.1	10.0.5.1
Last Valid Host	10.0.5.254	10.0.5.254
Total Valid Hosts	254	254

- **Observation**
  - The online subnet calculator displays the same network details as the manual calculation.
- **Result**
  - The manually calculated network address, broadcast address, CIDR notation, and valid host range were verified using an online subnet calculator.